

ACADEMIC RESEARCH & IMPLEMENTATION

AI-Generated Face Detection

A deep learning approach for facial authenticity verification leveraging pre-trained MobileNetV3-Large, RetinaFace alignment, and advanced regularization techniques.

Architecture: MobileNetV3-Large | Framework: PyTorch
Input Resolution: 224 x 224 x 3 | Best Performance: 99.06% Test Acc

Dataset representation & model quality:

- A deep learning model's face detection effectiveness is largely determined by training data diversity, size, and representation.
- A robust classifier must combine authentic human face images with synthetic images produced by state-of-the-art generators.
- Training on multiple generators ensures the neural network uncovers general facial irregularities instead of generator-specific patterns.
- This study incorporates real human images (FFHQ) and synthetic faces from two separate generators (Stable Diffusion & Nano Banana 2.0).

1. Real Faces (FFHQ)

- Source: Flickr-Faces-HQ dataset.
- Highly popular benchmark for face verification tasks.
- Significant diversity in age, gender, ethnicity, pose, and background.
- Reduces dataset bias.
- 15,000 real images selected to maintain training speed.

2. Stable Diffusion

- Represents modern diffusion-based generation.
- Significantly higher realism than early GAN networks.
- Challenges model with subtle synthetic irregularities.
- Forces detection of lighting, geometry, and noise inconsistencies.
- 9,001 AI-generated face images included in training corpus.

3. Nano Banana 2.0

- Evaluates cross-domain generalization capability.
- Overcomes dependency on Stable Diffusion anomalies.
- Merged directly into training, validation, and test splits.
- Significantly expands general face characteristics.
- Contributed 31,999 train, 4,000 val, and 4,000 test images.

Primary Dataset Construction

Target labels: Class 0 (Real Human Portrait) & Class 1 (AI-Generated Face).

Category	Number of Images
Real Images (FFHQ)	15,000
AI-generated Images (Stable Diffusion)	9,001
Primary Total	24,001

Final Consolidated Dataset

Initial stratified 80:10:10 split, merged with Nano Banana 2.0 partitions.

Dataset Split	Number of Images
Training Split	51,199
Validation Split	6,400
Testing Split	6,401
Final Combined Total	64,000

RetinaFace Alignment & Normalization

- Raw images vary in head pose, angle, illumination, background detail, and quality.
- A multi-stage pipeline is essential to remove variables unrelated to face authenticity.
- RetinaFace extracts face coordinates and maps 5 facial landmark positions (eyes, nose, mouth corners).
- Landmark alignment rotates and scales key details to static spatial positions.
- This reduces background bias and head tilt variables, forcing the classifier to study texture.
- Fallback center-crop is applied if the detector fails to identify facial coordinates.

Complete sequential pipeline mapping from Raw Input to Processed Output:

1. Face Detection

RetinaFace locates face and filters noisy pixels.

2. Landmark Alignment

Rotates coordinates based on eyes and nose corners.

3. Face Cropping

Removes backgrounds, haircut detail, and clothing shapes.

4. Spatial Resizing

Resizes to 224 x 224 resolution for MobileNetV3 compatibility.

5. Mean Normalization

Normalizes RGB channels based on ImageNet parameters.

6. Data Augmentation

On-the-fly transforms applied to training subsets only.

Spatial Resolution and Scale

- Crops the aligned facial box strictly to remove background details.
- Elements like landscapes, illumination sources, or outfits introduce training bias.
- Resizes crops to standard 224 x 224 dimensions.
- Preserves vital facial textures, iris shapes, and boundary outlines.
- Ensures uniform grid sizing for fast batch compilation and low GPU memory overhead.

ImageNet Input Normalization

Ensures input pixels follow pre-trained weights distributions.

Parameter	Red channel	Green channel	Blue channel
Mean (ImageNet)	0.485	0.456	0.406
Std Dev (ImageNet)	0.229	0.224	0.225

Advantages of standardization:

- Stabilizes model weight adjustments during early training.
- Ensures compatibility with ImageNet features, improving adaptation speed.

Geometric Transforms

- Random Resized Crop: Extracts different regions of the facial image.
- Improves robustness against face scale, positioning, and framing.
- Horizontal Flip: Flips images vertically with a 50% probability.
- Develops left-right orientation invariance, doubling dataset variants without label modifications.

Color & Blur

- Color Jitter: Modifies brightness, contrast, saturation, and hue.
- Simulates diverse illumination and camera configurations, preventing color-bias.
- Gaussian Blur: Blurs details with variable kernels.
- Simulates focal errors or motion blur, preventing reliance on high-frequency patterns.

Quality Degradations

- JPEG Compression: Artificially compresses image formats (JPEG formats).
- Prepares the model to identify deepfakes on social platforms after compression.
- Gaussian Noise: Perturbs individual pixel values.
- Simulates camera sensor noise, forcing the model to focus on structures instead of individual pixels.

Model Customization & Backbone

- The project utilizes Transfer Learning with the MobileNetV3-Large backbone, initialized with ImageNet pre-trained weights.
- MobileNetV3-Large offers an optimal balance between accuracy, parameter size, and latency.
- Its lightweight design supports easy deployment on edge devices and real-time platforms.
- The final ImageNet classification layer is replaced with a 2-output fully connected layer representing Real (0) vs AI-generated (1).
- All intermediate feature extraction blocks and internal layers are kept consistent.

Decoupled training stages prevent catastrophic loss of generic filters:

Stage 1: Feature Extraction (Epochs 1 - 3)

All parameters in the MobileNetV3-Large feature extraction backbone are frozen. Only the new classification layer is trained with a learning rate of 3×10^{-4} . This allows the classifier to learn high-level face shapes while preserving generic pre-trained visual representations.

Stage 2: Selective Fine-Tuning (Epochs 4 - 10)

The final 25% of MobileNetV3-Large feature blocks (blocks 12–16) are unfrozen and optimized with a smaller learning rate of 1×10^{-5} . This adapts the higher-level layers to identify subtle AI artifacts without destroying the low-level edge filters.

Runtime Environment Details

Component	Specification
Platform	Kaggle Notebooks
Framework	PyTorch
Computer Vision Library	Torchvision
Hardware Accelerator	Kaggle GPU Runtime
Mixed Precision Training	Automatic Mixed Precision (AMP)

Dataloader Parameters

Parameter	Value
Input Image Size	224 x 224 pixels
Number of Channels	3 (RGB)
Batch Size	128
Number of Classes	2 (Real, AI-generated)
Data Shuffling	Enabled for Training

1. Loss Function

- Uses standard CrossEntropyLoss.
- Directly compares predicted logits with ground truth binary classes.
- Provides stable gradients when paired with softmax outputs during model evaluations.

2. Optimizer (AdamW)

- AdamW selected over basic Adam.
- Decouples weight decay from the gradient update step.
- Weight Decay of 1×10^{-4} .
- Prevents weights from expanding excessively, improving general performance.
- Stage 1 LR: 3×10^{-4}
- Stage 2 LR: 1×10^{-5}

3. Cosine Scheduler

- Uses CosineAnnealingLR scheduler.
- Decreases learning rate smoothly following a half-cosine wave pattern.
- Enables large updates early, shifting to minor adjustments near the optimum.
- Avoids convergence oscillations.

Methodology & Setup

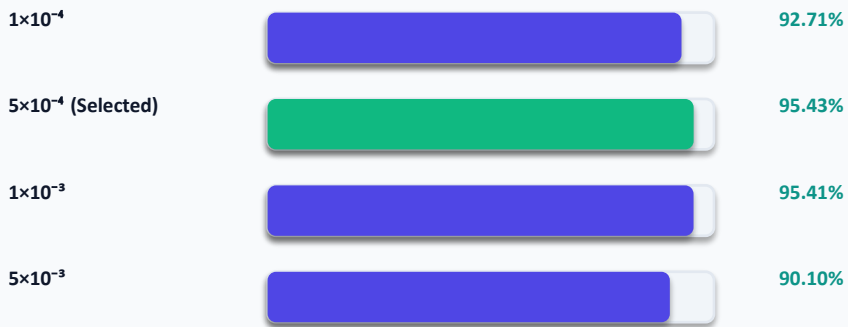
- Conducted a systematic study with 24 controlled ablation experiments.
- A single parameter is altered per experiment, keeping all other options static.
- Aims to optimize accuracy and F1-score while avoiding overfitting.
- Evaluates the impact of learning rate, batch size, regularization decay, dropout, optimization method, and label smoothing.

Evaluated Parameter Ranges

Hyperparameter	Tested Values
Learning Rate	1×10^{-4} , 5×10^{-4} , 1×10^{-3} , 5×10^{-3}
Batch Size	32, 64, 128
Weight Decay	0.0, 0.01, 0.05, 0.10
Dropout Rate	0.0, 0.2, 0.3, 0.5
Optimizer	Adam, AdamW
LR Scheduler	CosineAnnealingLR, ReduceLROnPlateau
Label Smoothing	0.0, 0.05, 0.10

Learning Rate (LR) Performance

Rates of 5×10^{-4} and 1×10^{-3} achieved optimal accuracy. High rates caused divergence, while low rates slowed adaptation.



Batch Size Performance

Batch size 32 offered marginal gains, but 128 was selected to maximize GPU acceleration.



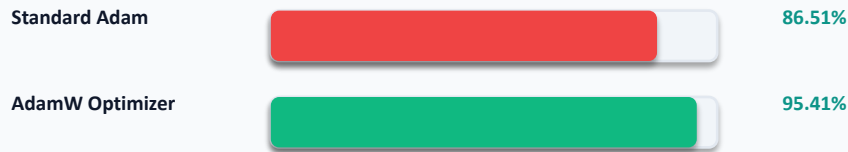
Dropout & Weight Decay

No additional dropout (0.0) worked best. Regularization from weight decay (0.05-0.10) improved F1 stability.



Adam vs AdamW Optimizer

AdamW's decoupled decay resulted in a 9% test accuracy improvement over standard Adam.



Learning Rate Scheduler

Cosine decay outperformed plateau-based reductions by adapting parameters more smoothly.

CosineAnnealing



95.41%

ReduceLROnPlateau



93.23%

Label Smoothing

Smoothing did not offer benefits, as the clean targets and dataset sizes minimized label noise.

No Smoothing (0.0)



95.41%

Smoothing 0.05



93.10%

Smoothing 0.10



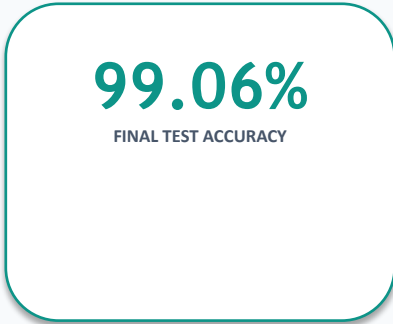
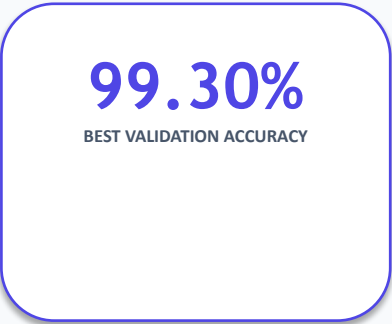
94.30%

Mitigating Overfitting & Optimizing Gradient Walks:

- Weight Decay (0.05 - 0.10): Penalizes large parameters, stabilizing high-level classification weights.
- Gradient Clipping: Limits gradient norms to 1.0. This prevents exploding gradients during stage 2 fine-tuning.
- Automatic Mixed Precision (AMP): Runs suitable matrix math in 16-bit float, reducing GPU memory footprint.
- On-the-Fly Transforms: Data augmentations (color jitter, blur, JPEG compression) act as regularization, preventing overfitting.
- Two-Stage Tuning: Selective freezing protects the lower-level edge and color filters learned from ImageNet.
- Cosine Decay Scheduler: Smoothly reduces the learning rate to settle into narrow local minima.

Core Performance Results

The optimized MobileNetV3-Large model achieved excellent classification performance, with minimal generalization gap between validation and test sets.



Classification Report

Class Label	Precision	Recall	F1-Score
Real Faces	0.9900	0.9929	0.9914
AI-generated Faces	0.9914	0.9879	0.9896

- Report insights:
- High precision and recall for both classes show balanced predictions.
 - A low generalization gap verifies the dataset was shuffled and split correctly.

Prediction Quality Overview

- Out of 6,401 total test samples, only 60 images were misclassified.
- Shows high classification reliability when detecting synthetic face irregularities.
- False Positives (25): Real human faces incorrectly classified as AI-generated.
- False Negatives (35): Synthetic AI faces incorrectly classified as real.
- The low false negative count shows model sensitivity to artificial textures.

Confusion Matrix

Actual / Predicted	Predicted Real	Predicted AI-Gen
Actual Real	3,475 (TN)	25 (FP)
Actual AI-Gen	35 (FN)	2,866 (TP)

Summary Conclusions

- High Accuracy: Reached a test accuracy of 99.06% and validation accuracy of 99.30%.
- Model generalizability: Cross-domain training (Nano Banana 2.0) successfully reduced generator bias.
- Efficient architecture: MobileNetV3-Large proved to be highly computationally efficient.

Future Work

- Test on newer models like Stable Diffusion XL or Midjourney v6.
- Incorporate frequency-domain analysis to detect subtle synthetic textures.
- Utilize explainable AI (Grad-CAM) to verify classification features.